

Techniques and algorithms checklist

Every implemented method, where it lives, why it is there, the formula, and what it must produce. No week labels — this is the software as built.

Read each row as a contract: **where** (module and file), **step and purpose**, **reason**, **formulae**, **expected output**. Module 2 rows 11–27 are the S21 teaching track plus the general MATLAB track and the Module 3 handoff. Module 3 and ROI start after that. Output is not a clinical diagnosis.

Module 1 — Data acquisition

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M1 · loader.py	1. Dispatch the file by extension	One entry point for the GUI and automation. Later modules never need to know the original format.	$\text{ext} = \text{lower}(\text{file suffix})$. .mat → MATLAB/BMID loader. .s1p/.s2p/.s4p/.s8p → Touchstone loader. Else <code>UnsupportedFileFormatError</code> .	Call to the correct parser.
M1 · validator.py	2. Check the path before parsing	Avoid crashing on a missing, empty or unknown file.	File must exist, size > 0, extension $\in \{.mat,.s1p,.s2p,.s4p,.s8p\}$.	OK to parse, or a typed error.
M1 · touchstone_loader.py	3. Import and inspect the Touchstone header; extract S21 on .s2p	Identify the frequency unit, data format and reference impedance. Imaging uses transmission S21, not the whole 2×2, on the UG track.	$f_{\text{Hz}} = f_{\text{GHz}} \times 10^9$; $f_{\text{Hz}} = f_{\text{MHz}} \times 10^6$; $f_{\text{Hz}} = f_{\text{kHz}} \times 10^3$. Option line: # UNIT S {RI MA DB} R Z0. Row order after f: S11, S21, S12, S22.	Header (unit, RI/MA/DB, Z0) and extracted S21 plus the full (n_f, n_ports, n_ports) matrix.
M1 · touchstone_loader.py	4. Convert S21 to complex linear form	Preserve magnitude and phase. Do not average, filter or subtract dB values.	DB: $A = 10^{(D/20)}$, $\phi = \phi_{\text{deg}} \cdot \pi/180$, $S21 = A(\cos\phi + j \sin\phi)$. MA: $S21 = \text{mag}(\cos\phi + j \sin\phi)$. RI: $S21 = R + jI$. $j^2 = -1$.	Raw complex S21 in linear units.
M1 · matlab_loader.py	5. Load legacy or v7.3 MATLAB files	Lab benches export both old MAT and HDF5 v7.3. Complex v7.3 arrays arrive as real/imag fields.	If <code>h5py.is_hdf5</code> : visit datasets and convert {real,imag} → real+j imag. Else <code>scipy.io.loadmat</code> . Drop keys starting with <code>__</code> .	Dictionary of numeric arrays.
M1 · matlab_loader.py	6. Find the frequency vector and S-parameter array	MATLAB files have no single schema.	Name hints: <code>freq/frequency/...</code> ; else longest real increasing vector. If $\max(f) < 1000$ treat as GHz: $f_{\text{Hz}} = f \times 10^9$. S: name hints <code>s_param/s21/...</code> plus dim matching n_f. Known cube fallback: <code>fd_data_s21</code> → 1–8 GHz, 1001 points.	f in Hz and S oriented with frequency on axis 0.
M1 · bmidi_loader.py	7. Slice one UM-BMID scan from the 3-D cube	A cube is many experiments. Loading without an index would mix them.	<code>cube[scan, freq, ant] ∈ ■</code> . $S = \text{cube}[k,:,:, \text{shape}(1001, n_{\text{ant}})]$, $f = \text{linspace}(1e9, 8e9, 1001)$. Pair <code>md_list_*</code> → labels <code>tum_diam/tum_rad, tum_x, tum_y, birads, ant_rad</code> .	One <code>MicrowaveDataset</code> plus tumor/healthy metadata in SI units.
M1 · physical_metadata.py	8. Parse radius, wave speed and FOV from comments	Reconstruction needs geometry even when it is only written in a note.	Regex for antenna radius (m/cm), wave speed (m/s), x span / y span. cm → m by /100.	metadata: <code>antenna_radius_m</code> , <code>wave_speed_m_per_s</code> , <code>reconstruction_*_span_m</code> .
M1 · validator.py · dataset_info.py	9. Validate frequencies and S; build <code>MicrowaveDataset</code>	Finite, non-empty arrays only. Downstream code uses one object for every format.	f finite, S finite, no NaN/Inf. $n_{\text{samples}} = S $. Summary: n_freq, range, n_ports, size, variable names.	<code>MicrowaveDataset</code> + <code>DatasetSummary</code> for the GUI panel.

Module 2 — Signal preprocessing and handoff

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M2 · preprocessing_pipeline.py	10. Route .s2p versus general / BMID	The UG brief is S21-only. BMID _adi is already cleaned and must not be cleaned again.	If Touchstone and .s2p → S21 track. Else five-stage general loop. BMID preset: every method = none.	Chosen track and a PreprocessingConfig.
M2 · touchstone_s21.py	11. Validate and sort samples	Prevent invalid values, duplicates and reverse frequency order from breaking IFFT.	$N_f = N_{S21}$. Sort so $f[k+1] > f[k]$. Drop/repair non-finite samples. Record notes.	Ascending frequency and S21 vectors of equal length.
M2 · touchstone_s21.py	12. Check uniform spacing; interpolate only if required	A conventional IFFT needs constant Δf . Matched target/reference must share one grid.	$\Delta f[k] = f[k+1] - f[k]$. If not uniform: $\Delta f_u = (f_{\max} - f_{\min}) / (N_u - 1)$, $f_u[k] = f_{\min} + k \Delta f_u$, $R_u = \text{Interp}(f, R, f_u)$, $I_u = \text{Interp}(f, I, f_u)$, $S_u = R_u + j I_u$. Linear interpolation on Re and Im.	Uniform f and complex S21. Flag interpolation_performed.
M2 · touchstone_s21.py	13. Build magnitude and phase representations	Inspect the original signal. Keep both wrapped and continuous phase.	$S_{\text{dB}} = 20 \log_{10} S $. $\phi_w = \text{atan2}(I, R)$. $\phi_u = \text{unwrap}(\phi_w)$. Unwrap fixes $\pm\pi$ jumps; it does not denoise.	Magnitude dB, Re, Im, wrapped phase, unwrapped phase (GUI plots).
M2 · touchstone_s21.py	14. Detect isolated invalid samples (Hampel / median / local)	Stop one glitch from becoming a broadband artefact after IFFT.	$m_k = \text{median}\{x[r]: r \in W_k\}$. $MAD_k = \text{median}\{ x[r] - m_k \}$. $\sigma_k = 1.4826 MAD_k$. Flag if $ x[k] - m_k > \lambda \sigma_k$ (λ commonly 3). Replace flagged Re/Im. Default method: hampel.	Corrected complex S21 and a count of repaired samples.
M2 · touchstone_s21.py	15. Average repeated sweeps when available	Reduce uncorrelated noise. Do not average different antenna positions or phantoms.	$S_{\text{m}}[k] = (1/M) \sum_m S_{21}^{(m)}[k]$. M = number of identical-condition repeats.	Complex-averaged S21, or skip with Averaging=No.
M2 · touchstone_s21.py	16. Optional reference subtraction in frequency	Remove a matched empty / tumor-free sweep when the user supplies one.	$S_{\text{diff}}[k] = S_T[k] - S_R[k]$. Same grid, same preprocessing.	Reference-subtracted S21, or skip.
M2 · filtering.py · touchstone_s21.py	17. Apply a mild noise filter on Re and Im	Reduce point-to-point noise without wiping a weak target bump. Never filter dB as if it were linear.	Default .s2p: Savitzky–Golay, window $L=7$, polyorder=3, on R and I. Also available: moving average $R_f[k] = (1/L) \sum R[k+r]$, Butterworth, median, Gaussian. $S_f = R_f + j I_f$.	Filtered complex S21. GUI comparison plots.
M2 · touchstone_s21.py	18. Quantify how much filtering moved the signal	Check that cleaning did not rewrite the measurement.	$\text{NRMSE} = \ S_f - S_{\text{before}}\ _2 / \ S_{\text{before}}\ _2$. Measures alteration, not image quality.	NRMSE in the validation report; chosen filter name and parameters.
M2 · week3_time_domain.py	19. Hamming window (keep unwindowed copy)	Taper band edges so IFFT sidelobes / ringing drop. Reconstruction still defaults to filtered S21(f).	$w_H[k] = 0.54 - 0.46 \cos(2\pi k / (N-1))$, $k=0 \dots N-1$. $S_H[k] = S_f[k] w_H[k]$.	S21_filtered and S21_Hamming both stored.
M2 · week3_time_domain.py	20. Window target and reference identically	Subtraction is only valid if both sides saw the same window.	$S_{\{T,H\}} = S_{\{T,f\}} w_H$ and $S_{\{R,H\}} = S_{\{R,f\}} w_H$. Same w_H , same N .	Matched windowed pair, or note that no reference was given.
M2 · ifft.py · week3_time_domain.py	21. IFFT to a complex time response	Delay-domain signals for clutter work and for DAS/DMAS.	$s[n] = (1/N) \sum_k S[k] e^{j 2\pi kn/N}$ (NumPy ifft). $\Delta t = 1/(N \Delta f)$, $t[n] = n \Delta t$, $T_{\text{period}} = 1/\Delta f$, $B = f_{\max} - f_{\min} = (N-1)\Delta f$. Need uniform Δf (rtol 1e-3).	time_s and time-domain traces with and without Hamming.
M2 · week3_time_domain.py	22. Matched reference subtraction in time	Suppress the tumor-free / system response. IFFT is linear so subtract-in-f or subtract-in-t is the same if windows match.	$s_{\text{diff}}[n] = s_T[n] - s_R[n]$ and $\text{IFFT}(S_T - S_R) = \text{IFFT}(S_T) - \text{IFFT}(S_R)$.	Reference-subtracted time (and frequency) copies when a reference exists.
M2 · week3_time_domain.py	23. Group-mean clutter removal, only if ≥ 2 channels	Subtract the response common to a set of comparable antennas. A single .s2p has $P=1$ so this is skipped.	$\mu_G[n] = (1/P) \sum_p s_p[n]$, $s_{\text{clean}}[n] = s_p[n] - \mu_G[n]$.	Clutter-reduced channels, or a skip note.
M2 · week3_time_domain.py	24. Optional global normalize	One scale for all channels. Do not normalize each channel on its own (that would destroy relative amplitudes).	$\alpha = \max_{\{p,n\}} s_{\text{clean}} $, $s_{\text{norm}} = s_{\text{clean}} / \alpha$.	Original-scale and globally normalized copies; α stored.
M2 · preprocessing_pipeline.py	25. General-track filter → calibrate → normalize → background → artifacts	Used for multi-trace MATLAB that is not a BMID pass-through.	Calibration: $S_{\text{cal}} = S_{\text{meas}} / S_{\text{ref}}$, or self-cal: subtract per-frequency mean across traces. Normalization (phase kept): $\max S $, $\min - \max S $, or z-score $ S $. Background: $S - S_{\text{bg}}$. Artifacts: hybrid / other methods in artifact_suppression.py.	Processed S of the same shape; stage_outputs snapshots.

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M2 · preprocessing_pipeline.py	26. Signal-quality report before vs after	Tell the GUI whether cleaning helped.	$SNR_{dB} = 10 \log_{10}(P_{smooth} / P_{residual})$ with 5-sample moving-average “signal”. $DR_{dB} = 20 \log_{10}(\max S /\min S)$. Artifact reduction = $1 - E_{after}/E_{before}$.	SignalQualityReport numbers on the Processing Summary panel.
M2 · handover_export.py	27. Validate and export the Module 3 handoff	Give reconstruction everything it needs, with traceability.	Check equal lengths, uniform f, finite complex values, matching metadata. Write .mat / optional .csv / .json.	frequency_Hz, time_s, S21_raw, S21_filtered, S21_Hamming, S21_reference_subtracted, S21_clutter_removed (if any), Tx/Rx coordinates, labels, processing_parameters, validation report.

Module 3 — Reconstruction

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
M3 · reconstruction_manager.py	28. Infer geometry (radius, c, FOV, array)	Pixels and delays are meaningless without a physical layout.	$x=\text{linspace}(x_{\text{min}},x_{\text{max}},n_x)$, $y=\text{linspace}(y_{\text{min}},y_{\text{max}},n_y)$, default 64×64. $a_i=(r \cos \theta_i, r \sin \theta_i)$. 360°: endpoint=False. 355° BMID arc: endpoint=True. Defaults: $r=0.08$ m, $c=3e8$, FOV ±5 cm. BMID: $r=0.18$ m, FOV ±6 cm. Optional $r_{\text{eff}}=0.97(r-0.106)+0.148$ m.	ReconstructionConfig and antenna positions ($N\times 2$ m).
M3 · ifft.py	29. Frequency → time for every antenna trace	Beamformers sample delays, not frequencies.	$\text{time_signals} = \text{ifft}(S, n=n_f+\text{pad}, \text{axis}=0)$. $t[n]=n/(N \Delta f)$. Fail if Δf is not uniform.	Complex ($N_{\text{time}}, n_{\text{traces}}$) array.
M3 · das.py	30. Delay-and-Sum (DAS)	Add each antenna’s envelope at the two-way delay of that pixel. Coherence weight fights ring clutter.	$d_i= a_i-r $, $\tau_i=2 d_i/c$, $n=\text{round}(\tau_i/\Delta t)$. $I_{\text{plain}}=\sum s_i(\tau_i) $. $CF=\sum s_i^2 / (N \sum s_i ^2)$. $I_{\text{DAS}}=I_{\text{plain}} \cdot CF^\gamma$, $\gamma=0.55$ ($\gamma=0$ is textbook DAS; cohort average).	One real image (n_y, n_x).
M3 · dmas.py	31. Delay-Multiply-and-Sum (DMAS)	Keep energy that is consistent on two antennas at once. Sharper than DAS, slower.	$p_{ij}=s_i \text{ conj}(s_j)$. $\text{signed_power}(z,p)=\text{Re}\{\text{sign}(z)(z +\epsilon)^p\}$, $p=0.30$, $\epsilon=1e-9$. $I_{\text{DMAS}}=\sum_{\{i<j\}} \text{signed_power}(p_{ij})$. Cohort average; textbook $p=0.50$.	One real image.
M3 · dmas_d4.py	32. DMAS-D4 contrast map	Compress dynamic range after DMAS. Cohort sweep keeps the textbook exponent 0.25. 0.55 was not better on average.	$I_{\text{D4}} = \text{sign}(I_{\text{DMAS}}) \cdot I_{\text{DMAS}} ^{\wedge\{0.25\}}$.	One real image.
M3 · quality/metrics.py	33. Image quality metrics	Compare the three pictures with numbers, not only by eye.	$\text{SNR}=\text{mean}(I)/\text{std}(I)$. $\text{SCR}=\text{max}(I)/\text{mean}(I-\text{mean}(I))$. $\text{CCR}=(\text{mean}(\text{top } 5\%)-\text{mean}(\text{rest}))/\text{std}(\text{rest})$. $\text{contrast}=(\text{max}-\text{min})/(\text{max}+\text{min})$. $\text{FWHM_px} = \text{mean of half-max widths on the peak row and column}$.	A metrics dict per image.
M3 · beamformer_selector.py	34. Select which beamformer to keep	One image goes to ROI. Mode depends on labels and the user.	Quality: $\text{score}=0.35 \text{ SNR}_n+0.35 \text{ SCR}_n+0.20 \text{ contrast}_n+0.10(1-\text{time}_n)$, metrics min–max normalized across the three. Also: prefer DMAS-D4, force DAS/DMAS/D4, or tumor_gt (closest ROI to labeled (x,y)).	Selected name + score table + reason string.
M3 · auto_calibrate.py	35. Auto Tweak: search geometry + beamformer	Wrong angle/flip/arc looks like a missed tumor. Search is only applied if it clearly beats the current settings.	Sweep angles {0.90,180,270}, flips, CW/CCW, 360/355°, phase-delay on/off, ROI modes. DAS first, then top-8 full beamformers. GT score= $1/(\text{dist}+1e-4)$. Apply if GT error drops by ≥0.5 mm.	Best config, or keep the baseline.

ROI, quality labels, and automation

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
ROI · roi_detector.py	36. Normalize the selected image	Threshold 0.70 must mean the same thing on dim and bright images.	$I_n=(I-I_{\text{min}})/(I_{\text{max}}-I_{\text{min}}+\epsilon)$, $\epsilon=1e-12$.	$I_n \approx [0,1]$.
ROI · roi_detector.py	37. Gaussian smooth	Stop a 1-pixel spike becoming its own region.	$I_s=\text{Gaussian}(I_n, \sigma=1.0 \text{ px})$.	Smoothed image.
ROI · roi_detector.py	38. Threshold + morphological opening	Turn intensity into candidate pixels, then drop speckle.	$B=1\{I_s\geq T\}$, $T=0.70$. Opening with a 3×3 ones kernel.	Binary map B.
ROI · roi_detector.py	39. Connected components	Separate islands are separate candidates.	Label connected 1s. Drop area < 6 px. If none: raise T by 0.05 (max 0.95) or box the global peak.	List of components / bounding boxes.
ROI · roi_detector.py	40. Score each component and pick spots	Prefer a compact interior hotspot, not the largest splash or an edge ring.	$\text{Score}=(0.65 I_{\text{peak}}+0.35 I_{\text{mean}})\times(A/A_{\text{bbox}})\times 1/(1+10 \text{ AreaRatio})\times \gamma_{\text{edge}}$, $\gamma_{\text{edge}}=0.65$ if the box touches the border else 1. Off-centre extra factor $(0.2+0.8 r_{\text{norm}})(1+1.5 r_{\text{norm}})$. Keep ≤4 spots, gap ≥6 px, score ≥0.40×best (auto 0.28). GT may add the nearest peak within 3 cm.	Ranked ROI list; primary = highest score.
ROI · reconstruction_manager.py	41. High-resolution rebuild inside the box	Same physics, finer pixels, smaller FOV.	Convert box to metres. $n=\text{clip}(\text{box}\times 4, \text{min } 96, \text{max } 256)$. Re-run DAS or DMAS or D4 on that span only.	ROIRefinement image + local spans.

Module / where	Step and purpose	Reason	Formulae and definitions	Expected output
ROI · characterization.py	42. Characterize the selected region	Describe size and focus in centimetres.	centroid via FOV interpolation. $d_{eq}=2\sqrt{(A_{cm}^2/\pi)}$. local SCR on the box. $FWHM_{cm}=FWHM_{px}\times px_{cm}$. compactness= $A_{px}/(w\ h)$. eccentricity= $(aspect-1)/(aspect+1)$.	TumorCharacterization numbers.
ROI · roi_detector.py	43. Tumor-candidate suspicion and YES/NO	Winning the ROI contest is not the same as looking tumor-like. Not a diagnosis.	local_contrast=peak/mean_outside. ring_excess=peak/median(same radius). Suspicion=clip($0.38 \tanh((lc-1)/2)+0.22/(1+(AR/0.08)^2)+0.15(0.45+0.55 r_{norm})+0.25 \tanh((ring-1.15)/0.35)$,0,1). Conf=clip($Susp\times 0.85_{edge}\times 0.65_{big}\times 0.55_{lowring}$,0,1). YES if Conf ≥ 0.45 . Forced NO if ring<1.40 (no close GT). GT ≤ 3 cm can force YES.	is_tumor_candidate, confidence, reason sentence.
ROI · confidence.py	44. Overall reconstruction confidence	One 0–1 number that blends picture quality, detection, beamformer margin, localization and focus.	overall= $0.22 image_q+0.28 detection+0.15 margin+0.22 localization+0.13 focus$. High ≥ 0.75 , medium ≥ 0.45 , else low. localization= $e^{-\{ -d_{cm}/2.5 \}}$ if GT exists.	ReconstructionConfidence + reasons.
Quality · tumor_taxonomy.py	45. Metadata taxonomy (labels only)	File experiments as healthy / small / medium / large, quadrant, shape. Not inferred from the image.	small if $d\leq 2$ cm, medium if $d\leq 4$ cm, else large. Quadrant from (x,y); central if $\sqrt{(x^2+y^2)}<1$ cm. Severity placeholder from size and BIRADS.	short_label such as small/moderate/upper-left or healthy.
Auto · discovery.py	46. Discover measurement vs metadata files	Do not treat md_list_*.mat as S-parameters.	fd_data_*.mat = cube. md_list_*/metadata_*.mat = labels. .sNp / other .mat = measurement.	List of DiscoveredFile with a kind and reason.
Auto · resolver.py	47. Pair files and pick BMID scans	A demo should cover healthy plus size/location variety, not 2000 identical clicks.	Strategies: diverse (default ~8), all, tumor_and_healthy, first. Tumor rank prefers larger d near ~3 cm offset.	ResolvedTarget list (path + scan index).
Auto · pipeline.py	48. Run the same M1→M2→M3→ROI chain unattended	Repeatable experiment. BMID: pass-through, FOV ± 6 cm, 64×64. Else: savgol 7/3 + time-domain stages on.	load → preprocess → reconstruct_all → select_best_beamformer → detect_rois → refine → characterize → confidence.	Per-file evidence dict + figures.
Auto · policy.py · classifier.py	49. Threshold profiles and pass/warn/fail	Mentors can tighten or loosen gates without editing physics.	Hard fail: load error, non-finite image, energy<min, ROI area<min. Soft warn: loc error, SCR, SNR, confidence, GT-but-not-candidate. balanced: loc ≤ 3 cm, SCR ≥ 2 , SNR ≥ 0.5 , Conf ≥ 0.35 , area ≥ 4 .	DatasetVerdict + MetricEvidence + recommended_upload.
Auto · reports.py · figures.py · cache.py · mcp_server.py	50. Write analysis pack; cache; MCP/CLI/GUI facade	One engine, three doors. Humans get markdown; machines get JSON.	Showcase = labeled tumor first, else strongest candidate. Files: latest_auto_analysis.md, latest_auto_validation.json, PNG copies.	Report paths and optional GUI handoff.

Required Module 3 handover (what export must contain)

Implemented in **preprocessing/handover_export.py**. The pack is: frequency_Hz and time_s; S21_raw, S21_filtered and S21_Hamming; S21_reference_subtracted and S21_clutter_removed where those stages ran; Tx_coordinates, Rx_coordinates and measurement_labels; processing_parameters and the validation report. If the file did not provide coordinates, Tx/Rx are placed on a circle of antenna_radius_m. Formats: .mat, optional .csv, .json.

50 rows. If a method is not in this table, it is not part of the implemented pipeline.